

Real-Time Control of Variable Air Volume System Based on a Robust Neural Network Assisted PI Controller

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Abstract—A neural network assisted proportional-plus-integral (PI) control strategy is proposed to improve the air pressure control performance of variable air volume (VAV) system. The neural network is trained online with a normalized training algorithm, which eliminates the requirement of a bounded regression signal to the system. To ensure the convergence of the training algorithm, an adaptive dead zone scheme is employed. Stability of the proposed control scheme is guaranteed based on the conic sector theory. To demonstrate the applicability of the proposed method, real-time tests were carried out on a pilot VAV air-conditioning system and good experimental results are obtained.

Index Terms—Convergence, neural networks, proportional-integral (PI) controller, stability proof.

I. INTRODUCTION

THE variable air volume (VAV) air-conditioning system can vary airflow rate to match the variation of space load and conserve the power of supply air fan at a reduced volume flow. The supply air pressure control is the key control task for VAV system. Its performance directly affects the whole system's energy efficiency. To model the dynamic behavior of the VAV system, some research work has been done [1], [2] based on physical principles. Mathematic models developed in these works reveal that high nonlinearity of the process is the main challenge for VAV control system design.

The traditional PI controllers have been used for heating, ventilating, and air-conditioning (HVAC) systems for a long time. However, owing to the existence of the nonlinear characteristics of system components, it is difficult to tune the controller parameters to achieve the maximum control performance. To help achieve an ideal operation performance, there have been a few interim solutions such as variable gain controllers [3], adaptive control algorithms [4]. As multiplayer neural networks can be considered as general tools for modelling nonlinear functions, in the recent years, research on neural-network-based adaptive control system has received a considerable attention [5]–[7]. The basic idea of applying neural networks to HVAC control systems was studied in [8]–[11]. However, neural network training algorithms are sensitive to disturbances and may not

be able to obtain an optimal performance without guaranteed stability.

This paper presents a neural network assisted PI (NNAPI) control scheme which is capable of maintaining good control performance under the situation that the actual cooling load is time-varying and system is highly nonlinear. We implement the design and stability analysis for this neural controller based on the conic sector theory with theoretical proofs. Real-time experiments were carried out on an experimental VAV system to control the static supply air pressure, and good experimental results were obtained.

II. NNAPI CONTROL SCHEME

Considering the dynamical control systems, which can be represented as a single-input-single-output (SISO) form as the following:

$$y(k) = f[(y(k-1), y(k-2), \dots, y(k-m_y), u(k-d-1), u(k-d-2), \dots, u(k-d-m_u)] + u(k-d) + \varepsilon(k) \quad (1)$$

where y is the output, u is the control input, d is the relative degree of the system. For the supply air pressure control loop, y is the static pressure in the supply air duct, and is the input voltage of supply air fan motor. $f(\cdot)$ is a unknown dynamic nonlinear function, $\varepsilon(k)$ demotes a bounded overall noise signal of the control system. m_y, m_u are integers and $m_y \geq m_u$. The set-point tracking error is

$$s(k) = y(k) - y^*(k) \quad (2)$$

where $y^*(k)$ is the command input. Define the control signal as

$$u(k-d) = -\hat{f}(\cdot) + y^*(k) + G_{PI}(z^{-1})s(k-1) \quad (3)$$

where $\hat{f}(\cdot)$ is estimate of $f(\cdot)$ by the neural network to be defined later, $G_{PI}(z^{-1})$ denotes the PI control algorithm. We have the control signal at time step k as

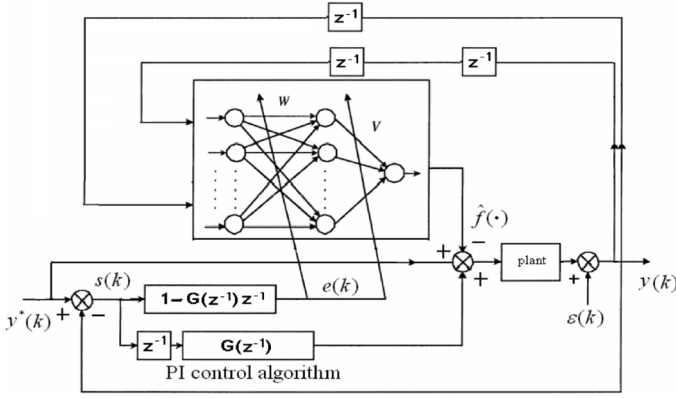
$$\begin{aligned} u(k) = & -\hat{f}(\cdot) + y^*(k+d) + G_{PI}(z^{-1})s(k-1+d) \\ = & -\hat{f}[y(k+d-1), y(k+d-2), \dots, y(k+d-m_y), \\ & u(k-1), u(k-2), \dots, u(k-m_u)] \\ & + y^*(k+d) + G_{PI}(z^{-1})s(k-1+d). \end{aligned} \quad (4)$$

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 Fig. 1. Structure of the control scheme (with $m_y = 2$, $m_u = 0$, $d = 2$).

For the fixed PI controller, we employ the discrete time algorithm

$$G_{PI}(z^{-1}) = k_P + k_I \cdot \frac{Tz^{-1}}{1 - z^{-1}}$$

where z^{-1} is the time delay factor, k_P is the proportional gain, k_I is the integral gain and T is the sampling time. To simplify the PI algorithm above, let $K_P = k_P$ and $K_I = k_I T$, we have the discrete time PI controller as

$$G_{PI}(z^{-1}) = K_P + K_I \cdot \frac{z^{-1}}{1 - z^{-1}}. \quad (5)$$

The estimation error vector of the neural network can be presented as

$$e(k) = f(k) - \hat{f}(k) + \varepsilon(k). \quad (6)$$

The neural network control structure is shown in Fig. 1. For the neural network training, since the estimation error $e(k)$ cannot be directly measurable, we use the tracking error to generate it basing on the closed-loop relationship via (1), (2), (3), (6)

$$e(k) = [1 - z^{-1}G_{PI}(z^{-1})] s(k). \quad (7)$$

The output of a three-layered neural network can be presented as

$$\hat{f}(k) = v^T(k)H(w(k), x(k))$$

where the input vector $x(k) \in R^m$ of the neural network is

$$x(k) = [y(k-1), y(k-2), \dots, y(k-m_y), u(k-d-1), u(k-d-2), \dots, u(k-d-m_u)]^T \quad (8)$$

with $m = m_y + m_u$.

$v(k) = [v_1(k), v_2(k), \dots, v_{n_h}(k)]^T \in R^{n_h}$ is the weight vector of the output layer, n_h is the numbers of neurons in the hidden layers of the network. The adjustable weight matrix $w(k) \in R^{n_h \times m}$ of the hidden layer is

$$w(k) = [w_1(k), w_2(k), \dots, w_{n_h}(k)]^T$$

where $w_j(k) = [w_{j1}(k) w_{j2}(k) \dots w_{ji}(k) \dots w_{jm}(k)]^T \in R^m$ with $j = 1, 2, \dots, n_h$, w_{ji} is the adjustable weight from the i th neuron of the input layer to the j th neuron of the hidden layer, $H(w(k), x(k))$ is the nonlinear activation function row vector

$$H(w(k), x(k)) = [h(w_1^T(k)x(k)) \quad h(w_2^T(k)x(k)) \quad \dots \quad h(w_{n_h}^T(k)x(k))] \quad (9)$$

where $h(\cdot)$ is the nonlinear activation function

$$h(x) = 1/(1 + e^{-4\lambda x}). \quad (10)$$

$\lambda > 0$ is the so-called gain parameter of the threshold function.

III. ADAPTIVE DEAD ZONE TRAINING ALGORITHM FOR NEURAL NETWORK

To deal with the disturbance, an adaptive dead-zone function is defined as

$$g(e(k), \Delta_v(k)) = \begin{cases} \|e(k)\| - \Delta_v(k), & \text{if } \|e(k)\| > \Delta_v(k) \\ 0, & \text{if } \|e(k)\| \leq \Delta_v(k) \end{cases} \quad (11)$$

where $\Delta_v(k)$ is the size of the adaptive dead-zone to be estimated.

Using the cost function

$$E(k) = (1/2)e^2(k) \quad (12)$$

we can define a normalized gradient decent algorithm to update $w(k)$

$$\begin{aligned} w(k+1) &= w(k) - \frac{\partial E(k)}{\partial w(k)} \frac{\beta(k)}{2 \left[1 + (\|H'(k)v(k)\| \|x(k)\|)^2 \right]} \\ &= w(k) + \frac{\beta(k)H'(k)v(k)e(k)x^T(k)}{2 \left[1 + (\|H'(k)v(k)\| \|x(k)\|)^2 \right]} \end{aligned} \quad (13)$$

$$\Delta_w(k+1) = \Delta_w(k) + \frac{\beta(k)\|e(k)\|}{2 \left[1 + (\|H'(k)v(k)\| \|x(k)\|)^2 \right]} \quad (14)$$

where $\beta(k)$ is a variable learning rate based on the adaptive dead zone scheme defined as

$$\begin{aligned} \beta(k) &= \begin{cases} g(e(k), \Delta_w(k)) / \|e(k)\|, & \text{if } \|e(k)\| \frac{h'_{\min}}{\lambda} > \Delta_w(k) \\ 0, & \text{if } \|e(k)\| \frac{h'_{\min}}{\lambda} \leq \Delta_w(k) \end{cases} \\ H'(k) &= \text{diag}[h'_1(k), \dots, h'_{n_h}(k)] \\ &= \text{diag}[h'(w_1(k), x(k)), \dots, h'(w_{n_h}(k), x(k))] \\ &\in R^{n_h \times n_h} \end{aligned} \quad (15)$$

where $h'_i(k)$ is the first derivative of the threshold function in (10), and $h'_{\min}(k) = \min[h'_1(k) \dots h'_{n_h}(k)] > 0$ (assume that

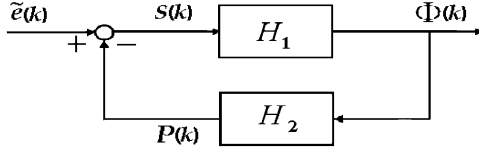


Fig. 2. Equivalent error feedback system.

$h'_i(k)$ is not simultaneously zero) is a positive constant. Similarly, a normalized gradient descent algorithm can be written in the matrix form

$$\begin{aligned} v(k+1) &= v(k) - \frac{\partial E(k)}{\partial v(k)} \frac{\alpha(k)}{1 + \|H(w(k), x(k))\|^2} \\ &= v(k) + \frac{\alpha(k) H(w(k), x(k)) e^T(k)}{1 + \|H(w(k), x(k))\|^2} \end{aligned} \quad (16)$$

$$\Delta_v(k+1) = \Delta_v(k) + \frac{\alpha(k) \|e(k)\|}{1 + \|H(w(k), x(k))\|^2} \quad (17)$$

where $\alpha(k)$ is a variable learning rate based on the dead zone scheme defined as

$$\alpha(k) = \begin{cases} g(e(k), \Delta_v(k)) / \|e(k)\|, & \text{if } \|e(k)\| > \Delta_v(k) \\ 0, & \text{if } \|e(k)\| \leq \Delta_v(k). \end{cases} \quad (18)$$

Refer to the structure of control scheme (see Fig. 1), the whole algorithm can be summarized as follows.

- 1) Form the new input vector $x(k)$ of the neural network based on (8).
 - 2) Calculate the neural network output $\hat{f}(k)$ using the input $x(k)$ and the existing weights.
 - 3) The control input $u(k)$ is calculated based on (3).
 - 4) The new plant output is taken and the new tracking error signal $s(k+1)$ is fed through a fixed filter to produce the training error signal $e(k)$ of the network according to (7).
 - 5) The training error $e(k)$ is used to train the neural network and calculates the new weights $w(k+1)$ and $v(k+1)$.
- Go back to step 1 to continue the iteration.

IV. STABILITY ANALYSIS

The stability analysis is based on the sector theory [15]–[17]. To apply the sector theory, the feedback system in Fig. 1 should be reformed as an equivalent feedback structure as shown in Fig. 2. Then neural network training algorithm can be proved to meet the certainty dissipative condition to obtain guaranteed stability of the tracking error signal $s(k)$.

The operator H_1 represents the adaptive dead zone training algorithm, its input signal is the tracking error $s(k)$ defined in (2). The output is $\Phi(k)$, which will be defined later and is related to the weight error vectors, and in turn, the estimation parameter error vector and tracking error $s(k)$ through (7). The operator H_2 usually represents the mismatched linear model uncertainty in a typical adaptive linear control system [15]. Here, the operator H_2 represents the fixed PI controller dynamics (refer to Fig. 1), i.e.,

$$H_2 = (1 - z^{-1}) / [1 - (1 + K_P)z^{-1} - (K_I - K_P)z^{-2}].$$

Now we can establish the relationship between the tracking error signal $s(k)$ and the parameter estimate vectors of the neural network

$$\begin{aligned} s(k) &= H_2 e(k) = H_2 (f(k) - \hat{f}(k) + \varepsilon(k)) \\ &= H_2 [v^T(k) \tilde{H}(k) + \tilde{v}^T(k) H(w^*, x(k)) + \varepsilon(k)] \\ &= H_2 [v^T(k) (H(w^*, x(k)) - H(w(k), x(k))) \\ &\quad + \tilde{v}^T(k) H(w^*, x(k)) + \varepsilon(k)] \\ &= H_2 [v^T(k) [h(w_1^{*T} x(k)) - h(w^T(k) x(k)) \dots \\ &\quad h(w_{n_h}^{*T} x(k)) - h(w_{n_h}^T(k) x(k))]^T \\ &\quad + \tilde{v}^T(k) H(w^*, x(k)) + \varepsilon(k)] \end{aligned} \quad (19)$$

where $\tilde{H}(k) = H(w^*, x(k)) - H(w(k), x(k))$, and $\tilde{v} = v^* - v(k) \in R^{n_h}$ is the weight error matrix of the output layer.

Note that the activation function $h(\cdot)$ in (10) is non-decreasing function. According to the mean value theorem, there exists a unique positive number $\mu(\cdot)$ with $\lambda \geq \mu(\cdot) \geq 0$, so (19) can be rewritten as the following:

$$\begin{aligned} s(k) &= H_2 \{v^T(k) [\mu(w_1^*, w_1(k), x(k)) (w_1^{*T} - w_1^T(k)) x(k) \\ &\quad \dots \mu(w_{n_h}^*, w_{n_h}(k), x(k)) \\ &\quad \times (w_{n_h}^{*T} - w_{n_h}^T(k)) x(k)]^T \\ &\quad + \tilde{v}^T(k) H(w^*, x(k)) + \varepsilon(k)]\} \\ &= H_2 v^T(k) \mu(k) \tilde{w}(k) x(k) \\ &\quad + H_2 [\tilde{v}^T(k) H(w^*, x(k)) + \varepsilon(k)]. \end{aligned} \quad (20)$$

Let

$$\Phi(k) = -v^T(k) \mu(k) \tilde{w}(k) x(k)$$

and

$$\tilde{e}(k) = H_2 [\tilde{v}^T(k) H(w^*, x(k)) + \varepsilon(k)]$$

which is an equivalent disturbance term.

We have

$$s(k) = -H_2 \Phi(k) + \tilde{e}(k). \quad (21)$$

Theorem 1: Consider the error feedback system (as shown in Fig. 2)

$$s(k) = \tilde{e}(k) - P(k)$$

$$\Phi(k) = H_1 s(k)$$

$$P(k) = H_2 \Phi(k)$$

with operators $H_1, H_2 : L_{2e} \rightarrow L_{2e}$ and discrete time signal $s(k), P(k), \Phi(k) \in L_{2e}$, and $\tilde{e}(k) \in L_2$. If the following holds:

- a) $H_1 : s(k) \rightarrow \Phi(k)$ satisfies

$$\sum_{k=0}^N [s^T(k) \Phi(k) + \sigma \|s(k)\|^2 / 2] \geq -\gamma \quad (22)$$

- b) $H_2 : \Phi(k) \rightarrow P(k)$ satisfies

$$\sum_{k=0}^N [\sigma P^T(k) P(k) / 2 - P^T(k) \Phi(k)] \leq -\eta \|(P(k), \Phi(k))\|_N^2 \quad (23)$$



Fig. 3. Measurement and control apparatus of VAV air-conditioning system.

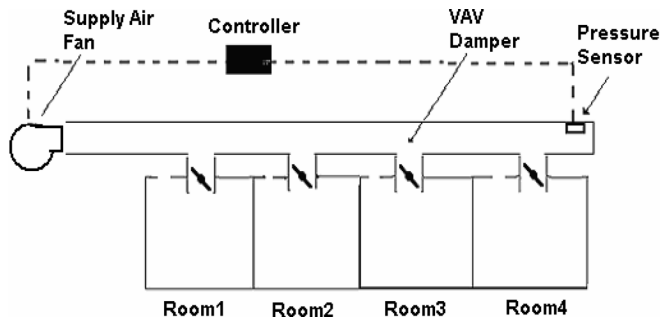


Fig. 4. VAV supply air pressure control loop.

for some $\sigma, \gamma, \eta > 0$, then the above feedback system is stable with $s(k), \Phi(k) \in L_2$.

Proof: See [17, Corollary 2.1].

Remark 1: It should be noted that the condition (b) can be treated as positive real function, i.e., the plot of H_2 should be in the positive half of a complex plane as shown in [16].

Lemma 1: The operator $H_1 : s(k) \rightarrow \Phi(k)$, which represents the training algorithm for hidden layer, satisfies condition (a) of Theorem 1.

Proof: See Appendix A.

Lemma 2: For the PI control algorithm employed in this controller design [refer to (5)], there exists one region, when the values of K_I, K_P are inside this region, the property of H_2 satisfy the strictly positive real (SPR) condition.

Proof: See Appendix B.

Therefore, both condition (a) and (b) of Theorem 1 are fulfilled, the control system is stable.

V. EXPERIMENTAL RESULTS

A VAV air-conditioning pilot plant was built for experimental purpose. As shown in Fig. 3, number 1, 2, 3, and 4 indicate the components of computer controller, air-conditioning pilot plant, signal process board and signal transmission cables, respectively. Fig. 4 is a simplified diagram of the supply air pressure control loop. The fan feeds fresh air into four rooms. Each room has one VAV box which can regulate the air volume by adjusting the damper's position. VAV dampers' position may change according to the cooling or heating requirement. The controller to be designed is supposed to maintain the supply air pressure at the set-point in spite of dampers' constant change. In

the real-time experiments, we compare the performance of the NNAPI controller with the conventional PI controller.

For a system with a static nonlinearity, the PI controller tuning procedure is difficult. One commonly used technique is to tune the controller when the plant is operating in the region of the highest gain so that the response may have damped performance at other operating points [9]. In our experiments, optimal PI controller's parameters were tuned with the well-known Ziegler–Nichols method under the condition that only one damper is completely open while others are closed, and the supply air pressure set-point is around 80 Pa. The proportional and integral gains for PI and NNAPI controller are obtained as $K_P = 1.5, k_I = 0.9$.

Experiment 1 is the performance comparison subject to different VAV damper status, for each controller, the tests were carried out two times with different VAV damper positions. For the first time, the dampers' position status: room 1 is completely opened, rooms 2–4 are completely closed and for the second time, the dampers' position status: room 1–4 are all completely opened. The experiment steps are described as follows: 1) stabilize the supply air static pressure at 60 Pa; 2) reset the supply air static pressure set-point to 80 Pa at the time step $t = 4000$; 3) reset the supply air static pressure set-point back to 60 Pa at the time step $t = 6000$. Figs. 5, 6, and Table I show the first experiment result. The PI controller was set around the operation points where the PI controller is well tuned before the damper position change. As we can see, after the change of damper positions (with more VAV dampers opened), the PI controller's performance degrades a lot in term of the response time. However, the NNAPI controller has overall superior performance compared to the standard PI controller in term of fast settling time.

Experiment 2 is the performance comparison of the two controllers under external disturbances subject to supply airflow rate change. The VAV dampers status: room 1 is completely closed, room 2–4 are completely open. These two different controllers go through the same test as follows: 1) stabilize the supply air static pressure at 40 Pa; 2) reset the supply air static pressure set-point to 60 Pa at the time step $t = 4000$; 3) close the damper of room 2 to generate a disturbance which will decrease the air flow rate in the duct at the time step $t = 3000$; 4) open the damper of room 2 again to generate a disturbance which will increase the air flow rate in the duct at the time step $t = 5000$; 5) reset the supply air static pressure set-point again to 40 Pa at the time step $t = 6000$. Fig. 7 shows the experiment 2 results. It is obvious that, with the same disturbance added in, the NNAPI controller has a faster settling time than the PI controller, which means that the disturbance rejection ability of this NNAPI controller is much stronger.

For the experiment 3, the integral gain of the well-tuned PI controller is decreased to simulate the PI control performance after system degradation due to the aging effect of long working hours. The proportional and integral gains are changed to $K_P = 1.5, K_I = 0.4$. The VAV damper status for the four rooms: room 1, 2 are completely open, room 3, 4 are completely closed. These two different controllers go through the same test as follows: 1) stabilize the supply air static pressure at 140 Pa; 2) begin at time step $t = 3000$, the set-point changes as a pulse signal, with period = 100 s and amplitude 15 Pa. From Fig. 8 we can

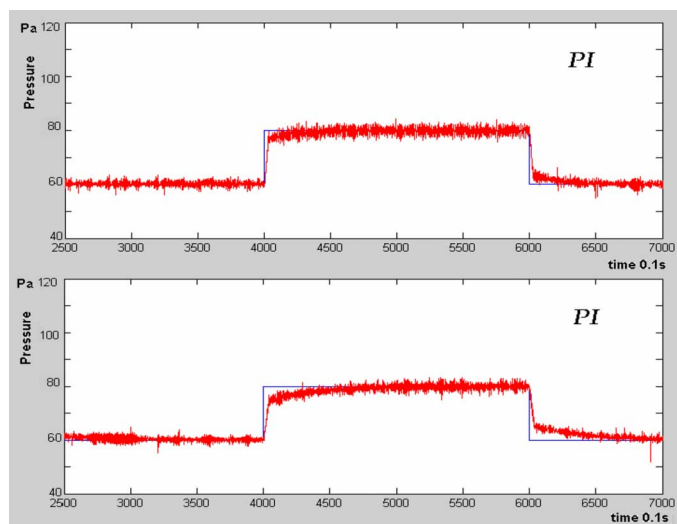


Fig. 5. Performance of PI controller under different VAV damper status in experiment 1. (Top) The damper of room 1 is completely open, those of room 2–4 are completely closed. (Bottom) The dampers of room 1–4 are all completely opened.

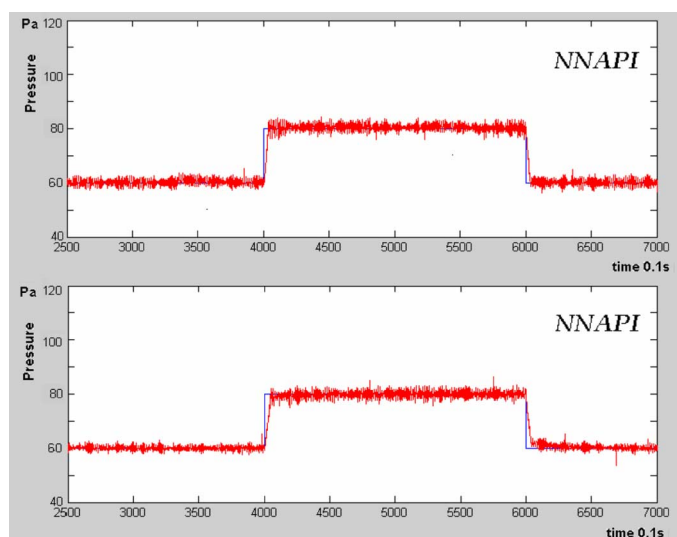


Fig. 6. Performance of NNAPI controller under different VAV damper status in experiment 1. (Top) The damper of room 1 is completely open, those of room 2–4 are completely closed. (Bottom) The dampers of room 1–4 are all completely opened.

TABLE I
PERFORMANCE COMPARISON FOR PI AND NNAPI
CONTROLLER IN EXPERIMENT 1

VAV dampers position	TESTING	CRITERIA	PI controller	NNAPI controller
One open, the other three closed	Set-point change from 60 Pa to 80 Pa	Settling time (s)	15	10
	Set-point change from 80 Pa to 60 Pa	Settling time (s)	25	12
All open	Set-point change from 60 Pa to 80 Pa	Settling time (s)	65	15
	Set-point change from 80 Pa to 60 Pa	Settling time (s)	70	20

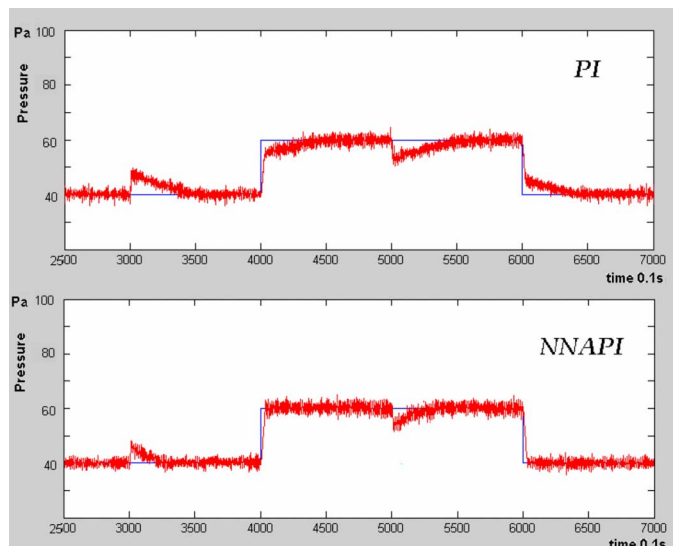


Fig. 7. Performance comparison for PI and NNAPI controller in experiment 2.

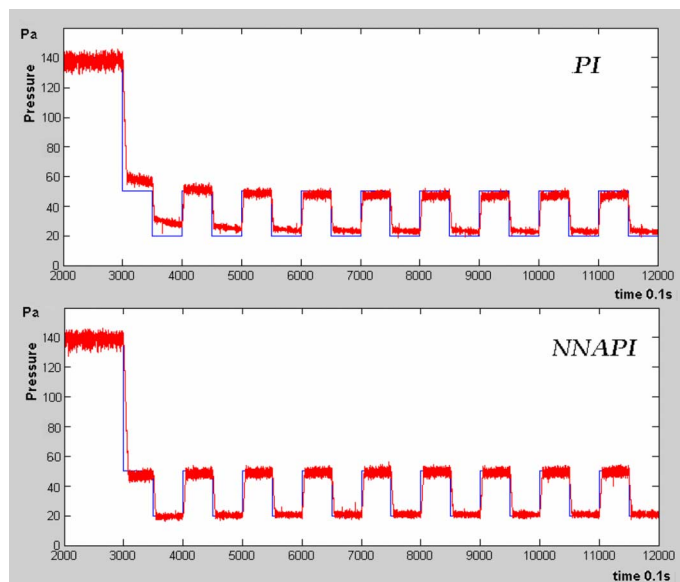


Fig. 8. Performance comparison for PI and NNAPI controller in experiment 3.

easily see that when the set-point changes from a higher level to a lower level, it takes the PI controller longer time to catch up with the set-point wave change. The NNAPI controller inherits the same parameters with the PI controller, which means, by integrating with the neural network trained by the adaptive dead zone train algorithm, the PI controller's performance degradation for aging effect is compensated to certain degrees.

VI. CONCLUSION

The focus of this paper is to design and demonstrate the robust performance of a neural network assisted PI control strategy for VAV air-conditioning system. The neural network is trained on-line to estimate the system nonlinear dynamic. To ensure the convergence of the learning algorithm and enhance the robust performance of the controller, an adaptive dead zone scheme is

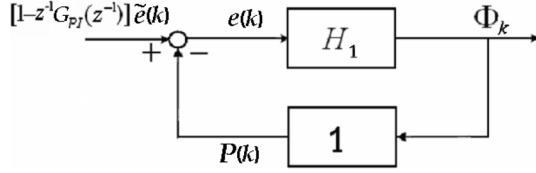


Fig. 9. Equivalent error feedback system.

used. The stability issue of the proposed control scheme is guaranteed based on the conic sector theory. Real-time experiments results demonstrate the applicability of the proposed method.

APPENDIX

A. Proof Of Lemma 1

First, we establish the conic sector condition between the estimate error $e(k)$ and the output $\Phi(k) = -v^T(k)\mu(k)\tilde{w}(k)x(k)$ (see Fig. 9) and then extend to the tracking error $s(k)$.

Define $V_w(k) = \text{tr}\{\tilde{w}^T(k)\tilde{w}(k)\}$, we have

$$V_w(k+1) - V_w(k) = -2\beta(k) \frac{\text{tr}\{H'(k)v(k)e(k)x^T(k)\tilde{w}^T(k)\}}{\rho(k)} + \beta(k)^2 \frac{\|H'(k)v(k)e(k)x^T(k)\|^2}{\rho^2(k)}. \quad (\text{A1})$$

Consider the property of the matrix trace, we have

$$\begin{aligned} & \text{tr}\{H'(k)v(k)e(k)x^T(k)\tilde{w}^T(k)\} \\ &= [h'_1(k)e^T(k)v_1(k)x^T(k)\tilde{w}^T(k) + \dots \\ & \quad + h'_{m_1}(k)e^T(k)v_{n_h}(k)x^T(k)\tilde{w}^T(k)] \\ &\geq \frac{h'_{\min}}{\lambda} \text{tr}\{e^T(k)v^T(k)\mu(k)\tilde{w}(k)x(k)\} \\ V_w(k+1) - V_w(k) &\leq -2\left(\frac{h'_{\min}}{\lambda}\right)\beta(k) \frac{\text{tr}\{e^T(k)v^T(k)\mu(k)\tilde{w}(k)x(k)\}}{\rho(k)} \\ &\quad + \beta(k)^2 \frac{\|H'(k)v(k)e(k)x^T(k)\|^2}{\rho^2(k)} \\ &= \beta(k) \left(2\frac{h'_{\min}}{\lambda}\rho^{-1}(k)e(k)\Phi(k) \right. \\ &\quad \left. + \rho^{-2}(k)\|H'(k)v(k)\|^2\|x(k)\|^2\beta(k)\|e(k)\|^2\right) \\ &= \beta(k) \left(2\frac{h'_{\min}}{\lambda}e^{nT}(k)\Phi^n(k) + \sigma(k)\|e^n(k)\|^2\right) \quad (\text{A2}) \end{aligned}$$

where $\sigma(k)\rho^{-1}(k)\|H'(k)v(k)\|^2\|x(k)\|^2\beta(k)$.

Summing the above equation upon to $N+1$ step, we are able to establish the conic condition with a constant $\hat{\sigma}$ for the estimate error $e(k)$ as the input conic sector theory (refer to Fig. 9) and the output $\Phi(k)$

$$\begin{aligned} & \sum_{k=0}^N \left[2\frac{h'_{\min}}{\lambda}e^{nT}(k)\Phi^n(k) + \rho^{-1}(k)\|H'(k)v(k)\|^2 \right. \\ & \quad \left. \times \|x(k)\|^2\beta(k)\|e^n(k)\|^2 \right] \geq -V_w(0) \\ & \sum_{k=0}^N \left[e^{nT}(k)\Phi^n(k) + \frac{\hat{\sigma}}{2}\|e^n(k)\|^2 \right] \geq -\frac{\lambda V_w(0)}{2h'_{\min}} \quad (\text{A3}) \end{aligned}$$

by selecting a suitable normalized factor $\hat{\rho}(k)$ to obtain the constant number $\hat{\sigma}$, such that $\hat{\sigma}$ satisfy $1 > \hat{\sigma} \geq \rho^{-1}(k)\|H'(k)v(k)\|^2\|x(k)\|^2\beta(k)\lambda/h'_{\min}$, and here we select $\hat{\rho}(k)$ as

$$\hat{\rho}(k) = 1 + \|H'(k)v(k)\|^2\|x(k)\|^2. \quad (\text{A4})$$

Note that the conic sector condition guarantees that the estimate error $e(k)$ and the output $\Phi(k) = -v^T(k)\mu(k)\tilde{w}(k)x(k)$ are bounded according the extended conic sector theory. And in turn, the parameter estimate error $\tilde{w}(k)$. The PI controller has the structure

$$G_{PI}(z^{-1}) = \frac{K_p + (K_I - K_P)z^{-1}}{1 - z^{-1}} \quad (\text{A5})$$

$$\begin{aligned} s(k) &= \frac{1 - z^{-1}}{1 - (1 + K_p)z^{-1} - (K_I - K_P)z^{-2}}e(k) \\ &= H_2e(k) \end{aligned}$$

$$e(k) = s(k) - [(1 + K_p)s(k-1) + (K_I - K_P) \times s(k-2) - e(k-1)]. \quad (\text{A6})$$

After substituting $e^n(k)$, (A2) can be rewritten a

$$\begin{aligned} & V_w(k+1) - V_w(k) \\ &\leq -2\left(\frac{h'_{\min}}{\lambda}\right)\beta(k) \frac{\text{tr}\{e^T(k)v^T(k)\mu(k)\tilde{w}(k)x(k)\}}{\rho(k)} \\ &\quad + \beta(k)^2 \frac{\|H'(k)v(k)e(k)x^T(k)\|^2}{\rho^2(k)} \\ &= 2\beta(k) \frac{h'_{\min}}{\rho(k)\lambda} \\ &\quad \times \{s(k) - [(1 + K_p)s(k-1) + (K_I - K_P)s(k-2) - e(k-1)]\}\Phi(k) \\ &\quad + \beta(k)\sigma(k)\rho^{-1}(k)\|e(k)\|^2 \\ &\leq 2\beta(k) \frac{h'_{\min}}{\rho(k)\lambda} s(k)\Phi(k) + \beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \\ &\quad \times \|(1 + K_p)s(k-1) + (K_I - K_P)s(k-2) - e(k-1)\|^2 \\ &\quad + \beta(k) \frac{h'_{\min}}{\lambda} \Phi^2(k) + \beta(k)\sigma(k)\rho^{-1}(k)\|e(k)\|^2. \quad (\text{A7}) \end{aligned}$$

The above inequality is due to the fact

$$\begin{aligned} & \frac{2}{\rho(k)} [(1 + K_p)s(k-1) + (K_I - K_P)s(k-2) - e(k-1)]\Phi(k) \\ &\leq \frac{1}{\rho^2(k)} \|(1 + K_p)s(k-1) + (K_I - K_P)s(k-2) - e(k-1)\|^2 + \Phi^2(k). \end{aligned}$$

For the PI control algorithm, we have

$$(1 + K_p)s(k-1) + (K_I - K_P)s(k-2) - e(k-1) = s(k) - e(k)$$

and

$$\begin{aligned}
& V_w(k+1) - V_w(k) - \beta(k) \frac{h'_{\min}}{\lambda} \Phi^2(k) \\
& \leq 2\beta(k) \frac{h'_{\min}}{\rho(k)\lambda} s(k) \Phi(k) + \beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|s(k) - e(k)\|^2 \\
& \quad + \beta(k) \sigma(k) \rho^{-1}(k) \|e(k)\|^2 \\
& = 2\beta(k) \frac{h'_{\min}}{\rho(k)\lambda} s(k) \Phi(k) - 2\beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} s(k) e(k) \\
& \quad + \beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|s(k)\|^2 + \beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|e(k)\|^2 \\
& \quad + \beta(k) \sigma(k) \rho^{-1}(k) \|e(k)\|^2 \\
& \leq 2\beta(k) \frac{h'_{\min}}{\rho(k)\lambda} s(k) \Phi(k) + 2\beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|s(k)\|^2 \\
& \quad + 2\beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|e(k)\|^2 \\
& \quad + \beta(k) \sigma(k) \rho^{-1}(k) \|e(k)\|^2 \tag{A8}
\end{aligned}$$

$$\begin{aligned}
& V_w(k+1) - V_w(k) - \beta(k) \frac{h'_{\min}}{\lambda} \Phi^2(k) \\
& \quad - 2\beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|e(k)\|^2 - \beta(k) \sigma(k) \rho^{-1}(k) \|e(k)\|^2 \\
& \leq 2\beta(k) \frac{h'_{\min}}{\rho(k)\lambda} s(k) \Phi(k) \\
& \quad + 2\beta(k) \frac{h'_{\min}}{\rho^2(k)\lambda} \|s(k)\|^2 \frac{V_w(k+1) - V_w(k)}{2\beta(k) \frac{h'_{\min}}{\lambda}} - \frac{\Phi(k)^2}{2} \\
& \quad - \frac{\|e^n(k)\|^2}{\rho(k)} - \frac{\|H'(k)v(k)\|^2 \|x(k)\|^2 \beta(k) \|e^n(k)\|^2 \lambda}{2h'_{\min}\rho(k)} \\
& \leq s^n(k) \Phi^n(k) + \frac{2}{\rho(k)} \frac{\|s^n(k)\|^2}{2} \\
& = s^n(k) \Phi^n(k) + \sigma \frac{\|s^n(k)\|^2}{2} \tag{A9}
\end{aligned}$$

with the normalized tracking error $s_n(k) = s(k)/\sqrt{\rho(k)}$, and the normalized output

$$\Phi^n(k) = \Phi(k)/\sqrt{\rho(k)} = -v^T(k)\mu(k)\tilde{w}(k)x(k)/\sqrt{\rho(k)}$$

by choosing the normalized factor $\rho(k)$ to guarantee that there exists a constant σ such that $1 > \sigma \geq 2\rho^{-1}(k) > 0$ i.e. $\rho(k) > 2$.

We can select

$$\rho(k) = 2 \left(1 + \|H'(k)v(k)\|^2 \|x(k)\|^2 \right)$$

and sum the above inequality up to $N+1$ steps, then we have

$$\begin{aligned}
& \sum_{k=0}^N \left[s^n(k) \Phi^n(k) + \frac{\sigma}{2} \|s^n(k)\|^2 \right] \\
& \geq -\frac{\lambda V_w(0)}{2\beta(k)h'_{\min}} \\
& \quad - \sum_{k=0}^N \left\{ \frac{\Phi(k)^2}{2} + \frac{\|e^n(k)\|^2}{\rho(k)} \right. \\
& \quad \left. + \frac{\|H'(k)v(k)\|^2 \|x(k)\|^2 \beta(k) \|e^n(k)\|^2 \lambda}{2h'_{\min}\rho(k)} \right\}. \tag{A10}
\end{aligned}$$

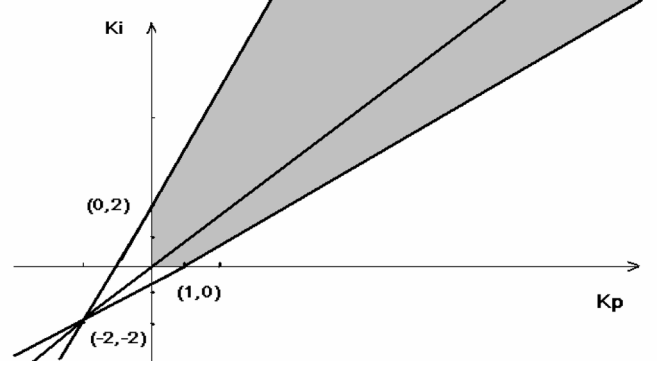


Fig. 10. Region in which the K_I, K_P can be selected to make the H_2 satisfy the SPR condition.

Remark 2: Note that the last item in the R.H.S of (A10) is bounded as shown in (A2) (obviously, we have $\rho(k) > \hat{\rho}(k)$, which implies that the normalization factor $\rho(k)$ will guarantee the conic sector condition. Furthermore, the specified normalized factor $\rho(k)$ plays two important roles. First, it guarantees $\sigma < 1$ to avoid the so-called vanished cone problem [17]. Second, it guarantees the sector conditions to be simultaneously satisfied by both the original feedback system in Fig. 1 and the equivalent feedback system in Fig. 2 [13], [16], [17]. This result satisfies condition (a) Theorem 1.

B. Proof of Lemma 2

$$\begin{aligned}
s(k) &= \frac{1 - z^{-1}}{1 - (1 + K_P)z^{-1} - (K_I - K_P)z^{-2}} e(k) \\
&= H_2 e(k). \tag{A11}
\end{aligned}$$

Note that $z = e^{j\omega} = \cos \omega + j \sin \omega$ and $\forall \omega \in [0, 2\pi)$, $i = \sqrt{-1}$.

We have

$$\begin{aligned}
H_2(e^{j\omega}) &= [(\cos 2\omega - \cos \omega) + j(\sin 2\omega - \sin \omega)] \\
& \quad / [(\cos 2\omega - (1 + K_P) \cos \omega + (K_P - K_I) \\
& \quad + j(\sin 2\omega - (1 + K_P) \sin \omega))] \tag{A12}
\end{aligned}$$

$$\begin{aligned}
\text{Re}[H_2(e^{j\omega})] &= \{2(K_P - K_I) \cos^2 \omega \\
& \quad - (2 + 2K_P - K_I) \cos \omega + 2 + K_I\} \\
& \quad / \{[\cos 2\omega - (1 + K_P) \cos \omega + (K_P - K_I)]^2 \\
& \quad + [\sin 2\omega - (1 + K_P) \sin \omega]^2\}. \tag{A13}
\end{aligned}$$

To make H_2 satisfy the SPR condition, i.e., $\text{Re}[H_2(e^{j\omega})] > 0 \forall \omega \in [0, 2\pi)$ we have

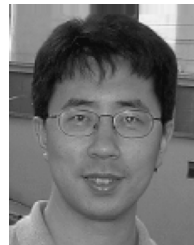
$$2(K_P - K_I) \cos^2 \omega - (2 + 2K_P - K_I) \cos \omega + 2 + K_I > 0 \tag{A14}$$

and note that $|\cos \omega| \leq 1$, and $K_P, K_I > 0$.

The above inequality can be solved as the result shown in Fig. 10's shadow area, which means that when the values of K_I, K_P is inside this region, the property of H_2 satisfy the strictly positive real condition (SPR).

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